Student Takeaway for Sync Session

<<Week 11: Paradigms of Machine Learning – I >>

This guide is your roadmap to making the most of our online session. Packed with essential tips and strategies, it's designed to keep you engaged, prepared, and ready to dive into a smooth and productive learning journey. Get ready to participate, learn, and thrive!

Session Overview

<table>
<tr><td>Session title</td><td>Paradigms of Machine Learning - I</td></tr>
<tr><td>Session duration</td><td>1 hour 45 minutes</td></tr>
<tr><td>Session type</td><td>Lectures: Exploration of key ML paradigms including supervised, unsupervised, semi-supervised, self-supervised, and active learning.
Use Cases: Real-world examples in classification, regression, imaging, and natural language tasks.
Framework Overview: Transfer, zero-shot, and few-shot learning.</td></tr>
<tr><td>Scope</td><td>This session explores core paradigms in machine learning and how they relate to real-world data scenarios. It covers:
Major learning types and their use cases
Comparative analysis of learning paradigms
Specialized techniques like transfer and zero-shot learning
Key tools and workflows in ML system design</td></tr>
<tr><td>Learning objectives</td><td><table><tr><th>Objective</th><th>Core capability</th></tr><tr><td>Understand key ML paradigms</td><td>Categorization of learning types</td></tr><tr><td>Distinguish learning approaches</td><td>Conceptual clarity with examples</td></tr><tr><td>Apply self-supervised and transfer learning concepts</td><td>Efficiency in model training</td></tr><tr><td>Recognize use cases of active learning</td><td>Data strategy awareness</td></tr></table></td></tr>
<tr><td>Software/Tools</td><td>Scikit-learn, TensorFlow, PyTorch
AutoML: H2O.ai
Monitoring: MLflow, Weights and Biases
Deployment: FastAPI, Docker, SageMaker</td></tr>
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Pro tips for success:

- **Compare Smartly**: Understand trade-offs between paradigms like supervised vs self-supervised
- **Apply Workflows**: Don't skip stages—use proper model selection, tuning, and monitoring
- **Stay Curious**: Think of new tasks where few-shot or transfer learning might be powerful
- **Experiment Freely**: Try deployment using Docker or track experiments via MLflow

Session Details

Slide(s)	Topic/Issue	What to expect	Notes
1-2	Paradigms of Machine Learning	Introduction to key paradigms of ML	Overview of various learning approaches
3	Types of Machine Learning: A Quick Recap	Recap of supervised and unsupervised learning	Classification and regression examples
4	Supervised Learning	Learning from labelled input-output pairs	Email spam detection, house price prediction
5	Unsupervised Learning	Discovering patterns in unlabelled data	K-means, PCA; used for segmentation
6	Semi-supervised Learning	Mixing labelled and unlabelled data	Cuts labelling costs, boosts model training
7	Semi-supervised Learning	Example of customer reviews classification	Workflow: train, predict, retrain
8	Self-Supervised Learning	Creating supervision from unlabelled data Process and transfer of knowledge	Uses pretext tasks like masked prediction Enables effective pretraining
9	Active Learning	Model selects uncertain samples for labelling	Best for costly/slow annotation scenarios
10	Transfer Learning	Pretrain on source, fine-tune on target	Saves time and reduces data needs
11	Zero-shot Learning	Predicts unseen classes using semantic info	Used in NLP and vision-language models
12	One-shot vs Few-shot Learning	Learning from minimal labelled examples	Used in Face ID, chatbots, signature checks
13	Warm-up Activity	Match each scenario to the correct learning type	Interactive exercise
14-15	Machine Learning Workflow	Key steps in building ML solutions	Lifecycle introduction

Slide(s)	Topic/Issue	What to expect	Notes
16	Step 1: Problem Definition	Goals, approach selection, success metrics	Align with business objectives
17	Step 2: Data Collection	Gathering relevant and reliable data	APIs, open datasets, internal logs
18	Step 3: Data Preparation and Exploration	Cleaning, feature engineering, splitting	Ready the data for training
19	Step 4: Model Selection	Comparing model types and trade-offs	Neural nets, decision trees, SVMs
20	Step 5: Model Training	Training procedures and dataset splits	80/20 split, avoid overfitting
21	Step 6: Model Evaluation and Tuning	Evaluation metrics and optimisation	F1, ROC-AUC, hyperparameter tuning
22	Step 7: Deployment and Monitoring	Ensuring model reliability in production	APIs, monitoring tools, retraining
23	Introduction to Machine Learning Tools	Key benefits for your workflow	Automate, collaborate, track performance
24	Problem Definition	Core objective and key tools	Miro, Notion, Lucidchart
25	Data Collection	Gathering raw data from multiple sources	Kafka, BeautifulSoup, SQL
26	Data Preparation and Exploration	Tools and tasks for data prep	pandas, NumPy, Seaborn, OpenRefine
27	Model Selection	Algorithms and tools	scikit-learn, XGBoost, AutoML
28	Model Training	Frameworks and tools	TensorFlow, PyTorch, GPU acceleration

Slide(s)	Topic/Issue	What to expect	Notes
29	Model Evaluation and Tuning	Key components and metrics	MLflow, Optuna, Weights & Biases
30	Deployment and Monitoring	Tools for scalable ML infrastructure	Docker, FastAPI, Kubernetes
31-33	ML Lifecycle Demo	Build and deploy a breast cancer classifier	End-to-end implementation
34	Q&A Session	Open floor for questions and clarifications	Summary and doubts

Post session

Reflection challenge	Which ML paradigm surprised you the most, and where can you apply it in real life?
Explore more	• **Read**: Research on zero-shot and few-shot learning in vision and NLP • **Watch**: Tutorials on MLflow, hyperparameter tuning with Optuna • **Experiment**: Train a model with minimal data using transfer learning
Get inspired	Many real-world AI systems use combinations of paradigms to succeed. Think about where you can blend them too—this is where ML innovation lives!
The journey ahead	Prepare for our next session on AI in Business Processes, where we'll examine how to integrate ML models into enterprise workflows for real impact.